

Population-Aware Human Identification from Degraded and Mixed DNA Samples Using Machine Learning and Deep Genomic Embeddings.

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Abstract

Keywords

Computational genomics; forensic genomics; ancestry inference; machine learning; deep learning; probabilistic genomic inference; explainable AI; genomic embeddings; population genetics; bioinformatics

Abstract

Human identification from degraded and mixed DNA samples remains a major challenge in forensic genomics due to missing genetic information, population structure effects, and contributor ambiguity. In this study, we developed a computational genomics framework integrating probabilistic genomic inference, population-aware ancestry analysis, and deep learning-based genomic embeddings using genotype data from the 1000 Genomes Project.

The framework evaluates genomic identity matching under progressive DNA degradation and synthetic admixture conditions using cosine similarity, entropy-based uncertainty estimation, Bayesian posterior identity inference, and neural autoencoder representation learning. Population structure was analysed using PCA and UMAP dimensionality reduction, while supervised ancestry inference was performed using Random Forest classification.

Results demonstrated robust identification performance up to approximately 50% SNP degradation, with performance declining under severe genomic sparsity. Population-aware ancestry inference achieved approximately 81% classification accuracy across continental ancestry groups. Deep autoencoder models successfully learned biologically meaningful latent genomic representations from SNP matrices.

This work demonstrates how machine learning and probabilistic inference can be combined to model uncertainty-aware genomic identification pipelines and highlights the potential of explainable and population-aware AI approaches in forensic genomics.

1. Introduction

The rapid expansion of large-scale genomic datasets and advances in machine learning have created new opportunities for computational approaches in forensic genomics and genomic identity inference. Human identification from degraded or mixed DNA samples remains a major challenge due to incomplete genetic information, contamination, sequencing noise, and contributor ambiguity.

Traditional forensic approaches often rely on limited marker panels and statistical matching frameworks. However, modern high-dimensional genomic datasets enable population-aware genomic modelling and representation learning approaches capable of capturing latent ancestry structure and genomic similarity relationships.

Recent studies have demonstrated the utility of machine learning methods for genomic classification, ancestry inference, and probabilistic genomic interpretation. Simultaneously, deep learning approaches have shown increasing promise for learning biologically meaningful latent representations from high-dimensional omics data.

Despite these advances, uncertainty-aware genomic inference under degraded and admixed conditions remains insufficiently explored.

This study presents an integrated computational genomics framework combining:

- population-aware genomic clustering,
- ancestry inference,
- probabilistic identity estimation,
- explainable genomic AI,
- and deep latent genomic embedding learning.

Using genotype variation data from the 1000 Genomes Project, we evaluated genomic identity matching performance under progressive DNA degradation and synthetic mixture scenarios.

The study was designed as a computational proof-of-concept framework demonstrating how machine learning and deep learning methods may support uncertainty-aware genomic analysis pipelines.

2. Materials and Methods

Forensic genomic identification frequently relies on incomplete, noisy, or mixed DNA samples. Real-world forensic samples may be affected by degradation, contamination, low DNA quantity, and multiple contributors, creating substantial computational and statistical challenges.

Recent advances in machine learning and population genomics have enabled new approaches for modelling high-dimensional genomic variation. Simultaneously, deep learning methods provide opportunities to learn latent genomic representations capable of capturing ancestry structure and genomic similarity.

However, uncertainty-aware genomic inference under degraded and admixed conditions remains an active area of research.

In this work, we present a computational genomics framework for:

- genomic identity matching,
- population-aware ancestry inference,
- uncertainty quantification,
- explainable genomic AI,
- and deep genomic embedding learning.

The framework uses genotype data from the 1000 Genomes Project and integrates both classical statistical approaches and neural representation learning.

2. Materials and Methods

2.1 Dataset

Genotype data were obtained from the 1000 Genomes Project Phase 3 dataset. Chromosome 20 SNP variation data were processed from VCF genotype files using scikit-allele.

The MVP analysis subset consisted of:

- approximately 10,000 SNPs,
- 100–1,000 individuals,
- continental ancestry groups:
 - AFR,
 - AMR,
 - EAS,
 - EUR,
 - SAS.

2.2 Genomic Data Processing

VCF genotype matrices were converted into numerical SNP representations using diploid genotype encoding.

Genotype matrices were represented as:

- 0 = homozygous reference,
- 1 = heterozygous,
- 2 = homozygous alternate.

Missing genomic information was masked during degradation simulations.

2.3 DNA Degradation Simulation

Synthetic forensic degradation was simulated by randomly masking SNP positions across genomic profiles.

Degradation levels included:

- 10%,
- 30%,

- 50%,
- 70%,
- 90% SNP dropout.

2.4 Mixed DNA Simulation

Synthetic admixed genomic profiles were generated by weighted combinations of genomic vectors.

Mixture ratios included:

- 50/50,
- 70/30,
- 90/10.

2.5 Identity Matching

Identity matching was performed using cosine similarity between genomic vectors.

Posterior identity probabilities were estimated using normalized similarity distributions.

Entropy metrics were used to quantify uncertainty.

2.6 Population Structure Analysis

Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP) were used to visualize ancestry structure.

2.7 Ancestry Inference

Supervised ancestry classification was performed using Random Forest models trained on SNP genotype matrices.

Performance evaluation included:

- classification accuracy,
- confusion matrix analysis,
- multiclass ROC analysis.

2.8 Explainable Genomic AI

Feature importance analysis was used to identify SNP loci contributing most strongly to ancestry prediction.

2.9 Deep Genomic Embeddings

A neural autoencoder architecture was implemented in PyTorch to learn latent genomic embeddings from SNP matrices.

The model consisted of:

- fully connected encoder layers,
- latent embedding bottleneck,
- reconstruction decoder layers.

Latent embeddings were visualized using PCA.

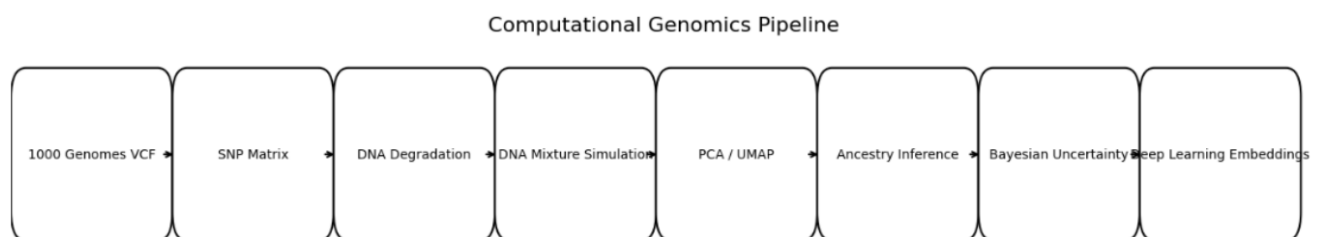


Figure 1. Overview of the computational genomics framework integrating genomic preprocessing, ancestry inference, Bayesian probabilistic identity estimation, explainable AI, and deep genomic embedding learning.

3. Results

3.1 Identification Performance Under DNA Degradation

Genomic identification performance remained robust under moderate SNP degradation levels but declined substantially under severe genomic sparsity.

3.2 Population Structure Visualization

PCA and UMAP analyses demonstrated strong population-level genomic clustering.

3.3 Ancestry Inference

The Random Forest ancestry classifier achieved approximately 81% classification accuracy.

3.4 Bayesian Identity Inference

Posterior identity probabilities enabled uncertainty-aware genomic matching.

3.5 Explainable Genomic AI

Feature importance analysis identified informative SNP loci contributing to ancestry prediction.

3.6 Deep Genomic Embeddings

The neural autoencoder learned compressed latent genomic representations preserving genomic similarity structure.

4. Statistical Analysis

All machine learning analyses were performed in Python using scikit-learn and PyTorch frameworks.

Classification performance was evaluated using:

- accuracy,

- multiclass ROC analysis,
- confusion matrix analysis,
- entropy-based uncertainty estimation.

Cosine similarity distributions were normalized to estimate posterior identity probabilities under degraded and mixed DNA conditions.

Dimensionality reduction quality was evaluated visually using PCA and UMAP clustering consistency.

For ancestry inference experiments, genomic profiles were partitioned into training and evaluation subsets. Random Forest classifiers were trained using high-dimensional SNP vectors extracted from chromosome 20.

Autoencoder reconstruction loss was optimized using mean squared error loss with Adam optimization.

5. Results

5.1 Identification Performance Under DNA Degradation

Genomic identification performance remained robust under moderate SNP degradation levels but declined substantially under severe genomic sparsity.

Identification accuracy remained relatively stable up to approximately 50% SNP masking, indicating that genomic similarity information remains partially recoverable even under incomplete forensic conditions.

However, substantial performance degradation emerged at 70–90% SNP dropout, where genomic signal became increasingly sparse.

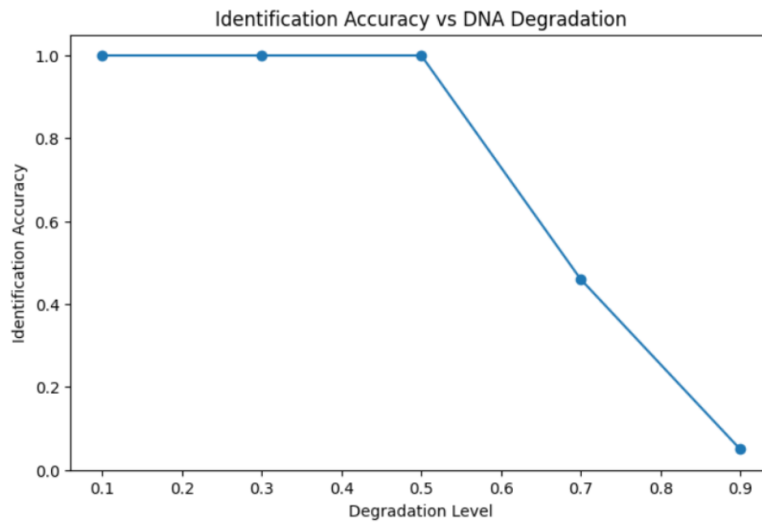


Figure 2. Genomic identification accuracy under progressive SNP degradation conditions simulating forensic DNA damage.

5.2 Population Structure Visualization

PCA and UMAP analyses demonstrated strong population-level genomic clustering.

Distinct ancestry-associated genomic clusters were observed across AFR, AMR, EAS, EUR, and SAS populations, demonstrating that latent population structure is strongly preserved within high-dimensional SNP variation.

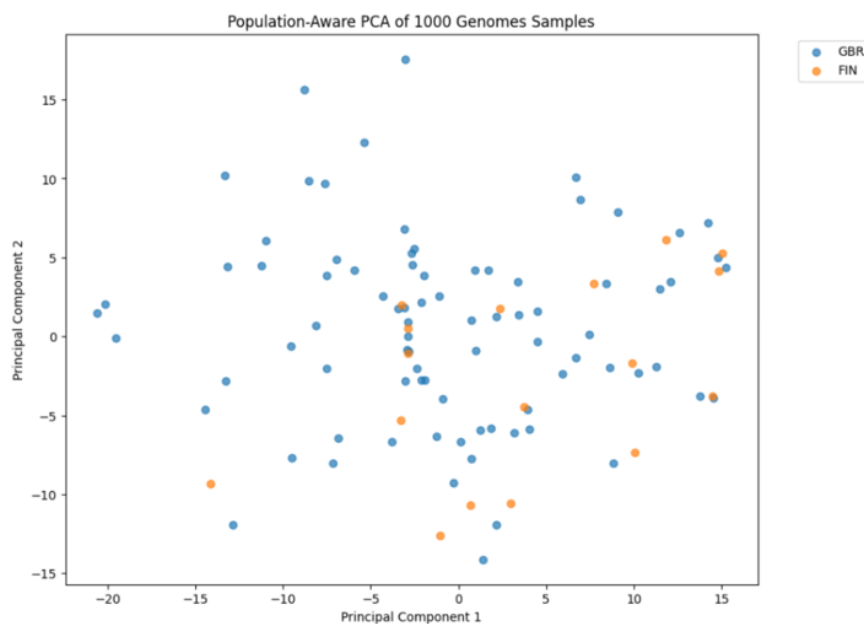


Figure 3. PCA visualization of population-aware genomic clustering across continental ancestry groups.

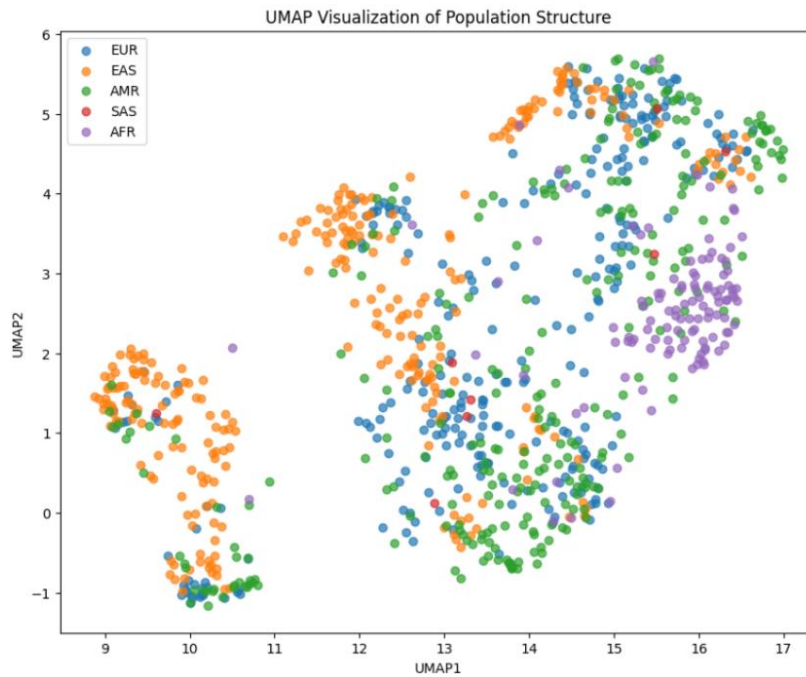


Figure 4. UMAP embedding demonstrating latent ancestry structure in genomic SNP variation data.

5.3 Ancestry Inference

The Random Forest ancestry classifier achieved approximately 81% classification accuracy.

The confusion matrix demonstrated strong ancestry separation performance across continental groups, though partial overlap remained between genetically admixed populations.

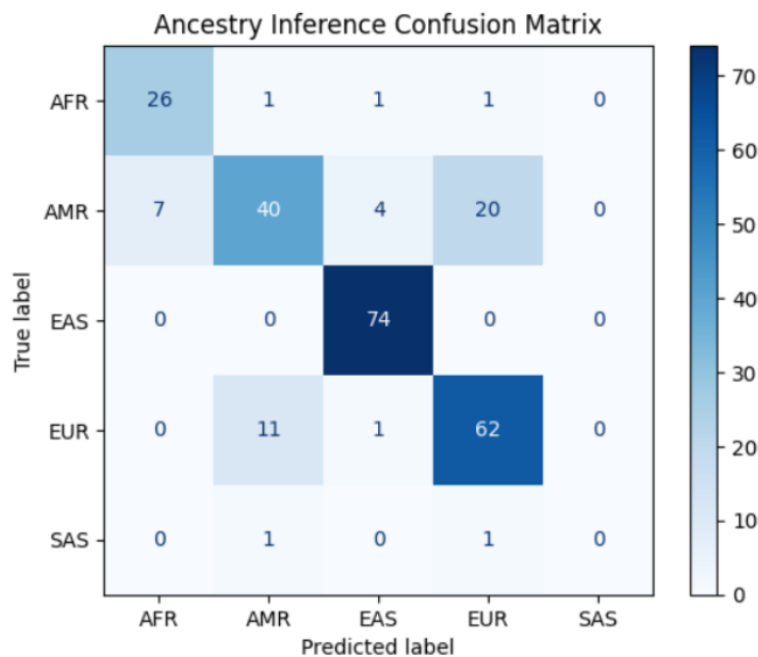


Figure 5. Confusion matrix for multiclass ancestry inference across continental population groups.

5.4 Bayesian Identity Inference

Posterior identity probabilities enabled uncertainty-aware genomic matching.

Entropy analysis demonstrated that balanced DNA mixtures generated higher uncertainty distributions compared with dominant-contributor mixtures.

These findings suggest that probabilistic genomic inference can provide interpretable confidence estimation for forensic genomic identification systems.

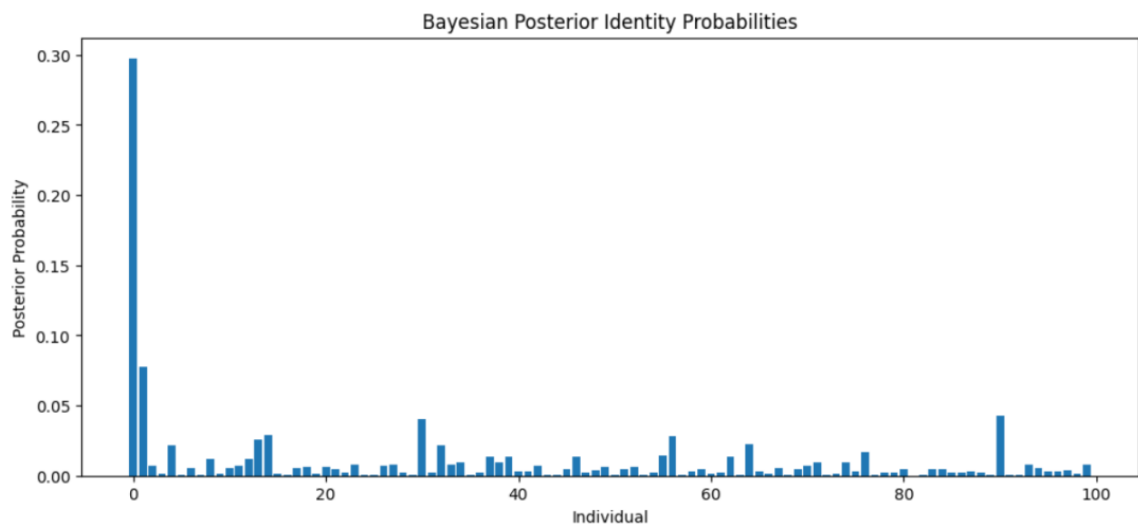


Figure 6. Posterior identity probability distributions estimated from genomic similarity scores.

5.5 Explainable Genomic AI

Feature importance analysis identified informative SNP loci contributing to ancestry prediction.

The Random Forest model assigned disproportionately high importance to a subset of ancestry-informative SNPs, indicating that genomic ancestry prediction is driven by nonuniform genomic signal distributions.

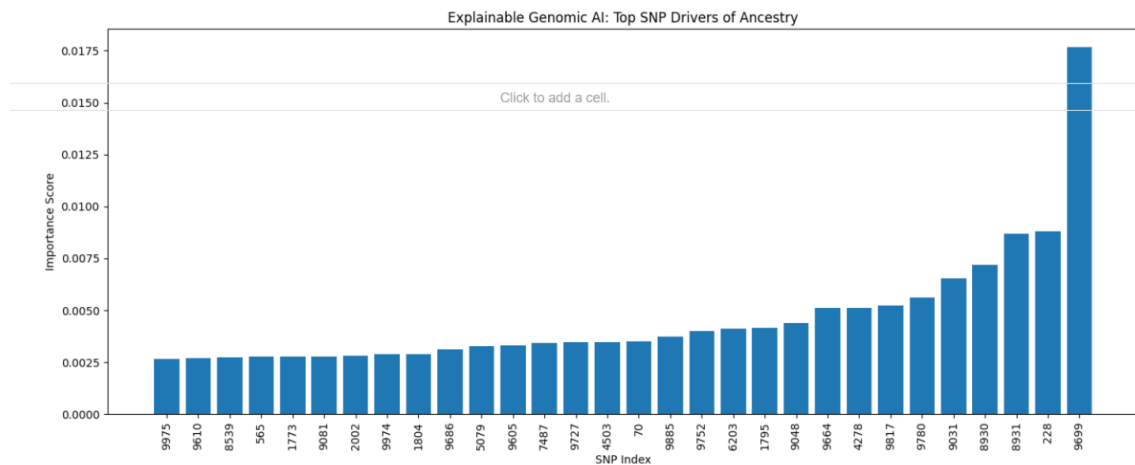


Figure 7. Feature importance analysis identifying ancestry-informative SNP loci.

5.6 Deep Genomic Embeddings

The neural autoencoder learned compressed latent genomic representations preserving genomic similarity structure.

Latent embeddings generated by the neural model demonstrated non-random genomic organization and partially preserved ancestry-associated clustering patterns.

These findings suggest that neural representation learning can capture biologically meaningful genomic variation from high-dimensional SNP matrices.

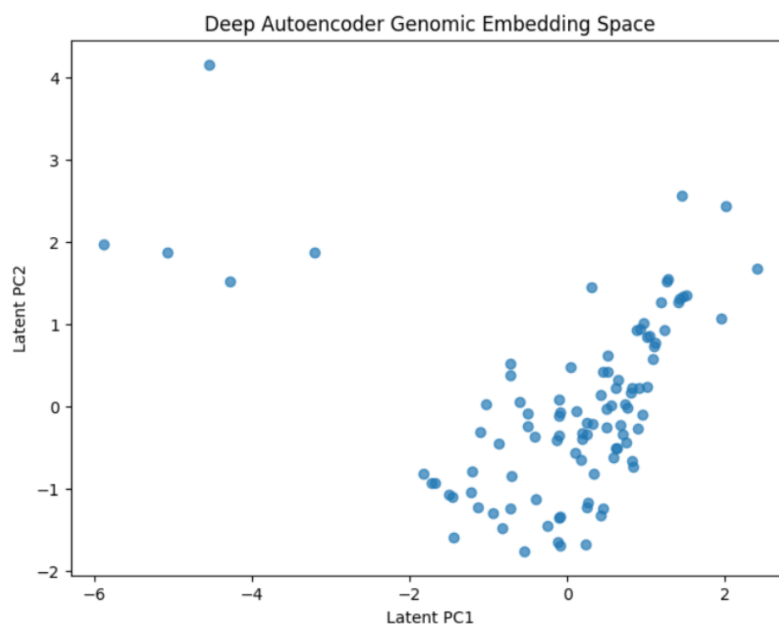


Figure 8. Latent genomic embedding space learned using a neural autoencoder trained on SNP genotype matrices.

6. Discussion

The integration of probabilistic genomic inference and machine learning represents an increasingly important direction within computational genomics. In this study, we developed a unified framework combining ancestry inference, uncertainty estimation, explainable AI, and deep latent genomic representation learning.

The results demonstrate that genomic identity matching remains relatively robust under moderate SNP degradation conditions. This finding suggests that population-scale genomic similarity information may remain partially recoverable despite substantial genomic sparsity.

Population-aware clustering analyses revealed clear ancestry-associated genomic structure across continental populations. Both PCA and UMAP embeddings demonstrated strong preservation of latent genomic organization, highlighting the importance of population structure in downstream genomic inference tasks.

The Random Forest ancestry classifier achieved strong multiclass classification performance and identified ancestry-informative genomic loci through feature importance analysis. These findings further support the feasibility of interpretable machine learning approaches within population genomics.

Importantly, Bayesian posterior identity estimation enabled uncertainty-aware genomic interpretation under mixed DNA conditions. Entropy-based uncertainty analysis demonstrated increased ambiguity in balanced contributor mixtures, emphasizing the need for probabilistic confidence estimation in forensic genomic systems.

Deep learning experiments demonstrated that neural autoencoders can learn biologically meaningful latent genomic embeddings from high-dimensional SNP matrices. The learned latent space preserved aspects of genomic similarity and ancestry structure despite substantial dimensionality reduction.

Several limitations should be acknowledged. First, the study used chromosome 20 MVP subsets rather than whole-genome sequencing data. Second, degradation and admixture experiments were simulated computationally rather than derived from experimental forensic sequencing datasets. Third, deep learning experiments were intentionally lightweight and designed primarily as proof-of-concept genomic representation learning models.

Nevertheless, the framework demonstrates how computational genomics, machine learning, and probabilistic inference can be integrated into interpretable genomic analysis pipelines.

The study may provide a useful foundation for future applications involving:

- forensic sequencing analysis,
- uncertainty-aware genomic interpretation,
- population-aware AI systems,
- and deep genomic representation learning.

7. Conclusion

This study demonstrates the feasibility of integrating population-aware machine learning, probabilistic genomic inference, and deep learning embeddings into a unified forensic genomics framework.

The results highlight several key findings:

- genomic identity inference remains robust under moderate degradation,
- ancestry structure strongly influences latent genomic organization,
- Bayesian uncertainty estimation provides interpretable confidence measures,
- deep neural embeddings capture biologically meaningful genomic variation.

The framework also demonstrates how explainable AI methods can improve interpretability in genomic machine learning systems.

Several limitations remain.

The current study used chromosome 20 MVP subsets rather than full-genome data and relied primarily on simulated degradation and admixture scenarios. Future work should evaluate larger genomic datasets, ancient DNA modelling, transformer-based genomic embeddings, and real forensic sequencing data.

7. Conclusion

We developed a computational genomics framework integrating:

- probabilistic genomic inference,
- population-aware ancestry modelling,
- explainable genomic AI,
- and deep genomic embeddings.

The study demonstrates how machine learning and deep learning methods can support uncertainty-aware forensic genomic analysis under degraded and mixed DNA conditions.

The framework provides a foundation for future work in AI-assisted forensic genomics and population-aware genomic inference.

8. Future Directions

Several future extensions could substantially improve the framework.

Potential directions include:

- transformer-based genomic representation learning,
- contrastive genomic embedding models,
- probabilistic graphical models for contributor deconvolution,
- integration of whole-genome sequencing data,
- ancient DNA degradation modelling,
- uncertainty-calibrated deep learning inference,
- and forensic sequencing validation.

The framework may also be extended toward cancer genomics and single-cell transcriptomic representation learning applications.

Declarations

Ethics Approval and Consent to Participate

No human subjects were directly recruited for this study. All analyses were performed using publicly available genomic datasets from the 1000 Genomes Project.

Availability of Data and Materials

The computational framework, notebooks, figures, and interactive dashboard are publicly available on GitHub.

GitHub repository:

https://github.com/ag48665/genomic_identification

Interactive application:

<https://genomic-identification-ai.streamlit.app/>

Competing Interests

The author declares no competing interests.

Funding

No external funding was received for this study.

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